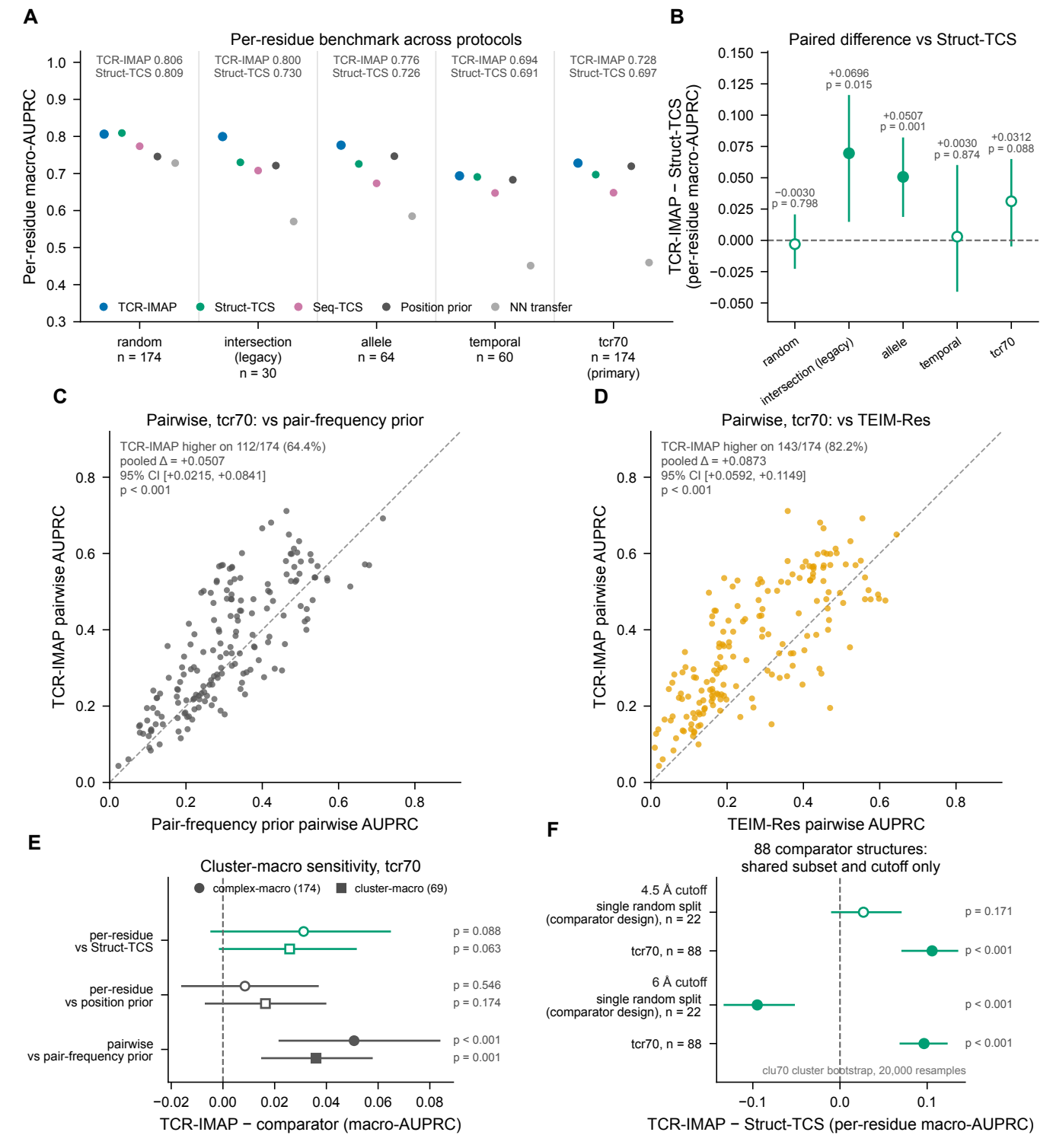
A
Per-residue benchmark across protocols
TCR-IMAP 0.806
Struct-TCS 0.809
TCR-IMAP 0.800
Struct-TCS 0.730
TCR-IMAP 0.776
Struct-TCS 0.726
TCR-IMAP 0.694
Struct-TCS 0.691
TCR-IMAP 0.728
Struct-TCS 0.697
Per-residue macro-AUPRC
TCR-IMAP
Struct-TCS
Seq-TCS
Position prior
NN transfer
random
n = 174
intersection
(legacy)
n = 30
allele
n = 64
temporal
n = 60
tcr70
n = 174
(primary)
B
Paired difference vs Struct-TCS
TCR-IMAP − Struct-TCS
(per-residue macro-AUPRC)
−0.0030
p = 0.798
+0.0696
p = 0.015
+0.0507
p = 0.001
+0.0030
p = 0.874
+0.0312
p = 0.088
random
intersection (legacy)
allele
temporal
tcr70
C
Pairwise, tcr70: vs pair-frequency prior
TCR-IMAP higher on 112/174 (64.4%)
pooled Δ = +0.0507
95% CI [+0.0215, +0.0841]
p < 0.001
TCR-IMAP pairwise AUPRC
Pair-frequency prior pairwise AUPRC
D
Pairwise, tcr70: vs TEIM-Res
TCR-IMAP higher on 143/174 (82.2%)
pooled Δ = +0.0873
95% CI [+0.0592, +0.1149]
p < 0.001
TCR-IMAP pairwise AUPRC
TEIM-Res pairwise AUPRC
E
Cluster-macro sensitivity, tcr70
complex-macro (174)
cluster-macro (69)
per-residue
vs Struct-TCS
p = 0.088
p = 0.063
per-residue
vs position prior
p = 0.546
p = 0.174
pairwise
vs pair-frequency prior
p < 0.001
p = 0.001
TCR-IMAP − comparator (macro-AUPRC)
F
88 comparator structures:
shared subset and cutoff only
4.5 Å cutoff
single random split
(comparator design), n = 22
p = 0.171
tcr70, n = 88
p < 0.001
6 Å cutoff
single random split
(comparator design), n = 22
p < 0.001
tcr70, n = 88
p < 0.001
clu70 cluster bootstrap, 20,000 resamples
TCR-IMAP − Struct-TCS (per-residue macro-AUPRC)